REU Research Projects

REU student selections are made with the goal of providing the 'best fit' between the projects offered by the faculty and the summer participant's background and interests.

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Summer 2026 Research Projects List

Advancing Decarbonization and Energy Performance of Concrete via 3D Printing —

Florence Sanchez, Civil and Environmental Engineering

The Sanchez lab focuses on the development of sustainable and more durable infrastructure materials with reduced environmental impact. One effort is the realization of novel, functionally graded concrete materials that can enhance the energy performance of buildings. This is achieved by engineering concrete at multiple length scales through the design of a hierarchy of internal structures inspired by nature and the incorporation of nano/micro-inclusions using extrusion-based 3D printing technology. The student will assist in the 3D printing fabrication of cement-based materials displaying a hierarchy of internal structures and patterns and investigations into the structure and mechanical and thermal performance of these novel materials. The student will gain firsthand experience in laboratory research, will be exposed to fundamental materials science and engineering of composite materials, and will develop skills in state-of-the-art analytical methods for the characterization of material microstructures.

Researcher Interests and Skills: This project is best suited for a student interested in material synthesis and characterization, 3D printing, and the applications of materials in civil engineering. Student must have completed at least one semester of chemistry and two semesters of a laboratory course.

Attosecond Quantum Dynamics at the Nanoscale —

Kalman Varga, Physics and Astronomy

Kalman Varga's group focuses on computational modeling and simulation of electronic structure and transport in nanoscale systems driven by intense, ultrashort laser pulses. Their research explores time-dependent electron dynamics, including Coulomb explosion, petahertz electronics, attochemistry, ultrafast band-structure control, and rapid energy-transfer mechanisms. The team also investigates cavity quantum electrodynamics, examining how quantized light interacts with matter.

Researcher Interests and Skills: We welcome students interested in computational nanoscience. A basic background in quantum mechanics and programming is helpful, but motivated students will receive training to develop these skills.

Biohybrid Solar Cells that Exploit the Z Scheme of Photosynthesis —

G. Kane Jennings, Chemical and Biomolecular Engineering
David Cliffel, Chemistry

Photosynthesis is the Earth's 95 TW solar energy conversion process that dwarfs our rate of energy usage from fossil fuels by nearly an order of magnitude. The Photosystem I and II protein complexes (PSI and PSII) are the drivers of photosynthesis in green plants. These two proteins absorb mostly red and blue light and efficiently translate that light energy to a charge separation with over 1 Volt of potential difference across the thylakoid membrane. These two powerful light-triggered events energize the Z-scheme of photosynthesis for the ultimate production of NADPH. While the overall efficiency of photosynthesis is low, the internal quantum efficiencies of PSI and PSII approach 100%. As such, researchers have long sought to extract and incorporate either the PSI or PSII protein complexes with electrodes to fabricate biohybrid solar cells.

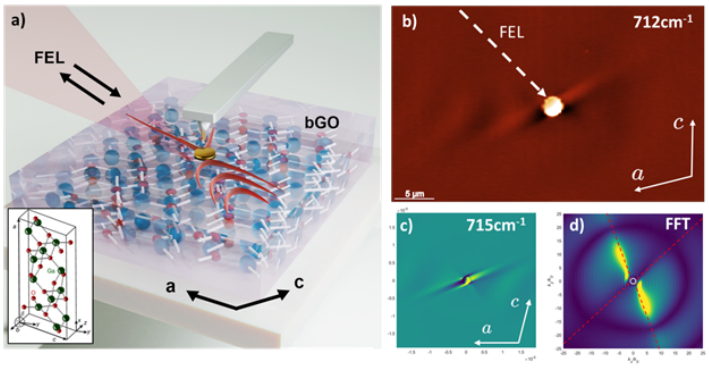
This project seeks to extract both PSI and PSII protein complexes from agricultural sources, modify them photoelectrochemically with metal nanoparticles or conducting polymers, and incorporate them both as the active elements in portable biohybrid solar cells. Thus, the project will introduce a sustainable approach to employ leafy agricultural byproducts for renewable energy. This project will combine PSI and PSII in the same pseudo-solid-state solar architecture to elevate cell potentials and recreate the Z scheme of photosynthesis within a biohybrid solar cell. The research addresses a particular challenge that has limited biohybrid solar cell performance—the wiring of the key reaction centers of PSI and PSII for direct electron transfer without resorting to expensive, yield-killing genetic modifications. This challenge will be surmounted by using the proteins' active sites to photooxidize monomer or photoreduce metal salts and generate conducting protein-polymer or protein-particle conjugates. The asymmetry of these conjugates will provide chemical handles to orient the particles on an electrode of high surface area, and insulative backfilling around the

proteins will greatly reduce charge recombination. The focus of this project is toward the materials interfaces and nanotechnology of these protein complexes.

Researcher Interests and Skills: This project is appropriate for a student who has an interest in investigating alternative energy and in characterizing materials and surfaces. Students should have completed general and organic chemistry courses.

Controlling Infrared Light Propagation at the Nanoscale with 2D van der Waals Crystals —

Josh Caldwell, Mechanical Engineering



The infrared portion of the electromagnetic spectrum offers opportunities for non-contact measurements of temperature through the collection of thermal radiation, chemical vibrational fingerprinting for sensing and identification, lab-on-a-chip, environmental and remote sensing, and astronomical observations. However, the long free-space wavelengths of light in this spectral domain restrict such optical components to large form-factors limiting the ubiquitous implementation in commercial applications. Furthermore, most of the materials used are either opaque in the visible, hygroscopic, or only work over a very narrow band, while detectors typically require cryogenic cooling for efficient operation.

The field of nanophotonics seeks to identify schemes by which the wavelength can be compressed to deeply sub-wavelength dimensions, opening the door to implementation of more traditional semiconductor and dielectric materials, overcoming many of these challenges. However, despite this promise, we continue to uncover additional materials featuring novel optical properties. One specific class is highly anisotropic (low symmetry) crystals, such as van der Waals or so-called two-dimensional materials or wide bandgap semiconductors. One example is through confining and propagating infrared light through hyperbolic shear polaritons in monoclinic crystals, which results in highly directional propagation of light, with the direction tunable through changing excitation frequency (see figure).

An REU student working on this project will implement these materials for demonstrating such control of IR light propagation, using state-of-the-art 2D material transfer and lithography tools, along with learning to use advanced IR spectroscopic methods including Fourier transform infrared (FTIR) spectroscopy and microscopy, and the use of nano-optic probes such as scattering-type scanning near-field optical microscopy (s-SNOM) and nano-FTIR. To date, all seven undergraduate

researchers (four of which were VINSE REU student alumni) of the Caldwell lab (since joining faculty in 2017) have seen their work published in a peer-reviewed journals, with this past summer's REU student on track for a publication this Fall.

Researcher Interests and Skills: This project would be ideal for students interested in practical research in nanomaterial fabrication and optical characterization techniques. Ideally students should have completed at least one course covering basic optics, electromagnetics, solid-state physics/chemistry, materials science, or the optical properties of materials. Undergraduate research, especially in materials science, optics, spectroscopy, or experience with cleanroom use, are highly desired.

Covalent Bonding of Hematite to Silica Surfaces —

Janet Macdonald, Chemistry

The hematite (iron oxide) rock paintings of the Anishinaabe of the Northern US and Canada are often on exposed rock faces that suffer extreme weathering conditions. Spray paint graffiti on the same sites can be worn off within a year or two, yet the rock paintings persist. Unfortunately, the methods to make such paintings have been lost. What is the chemistry behind their indelibility? Can we use our knowledge of chemistry to re-discover the painting technique? Can lab reagents and conditions be translated to "natural" granite surfaces and naturally sourced materials? The researcher will develop experiments to test hypotheses put forward previously by anthropologists and by chemists (us!), by making their own pigments, varying the chemical formulation of the paint and testing their durability.

Researcher Interests and Skills: This project requires a minimum of senior level high school, AP or freshman chemistry and a strong interest in Chemistry or Materials Science. Students should have a secondary interest in Anthropology, History or Indigenous Studies. Given the nature of the project, interested researchers should be prepared to be respectful of cultures and religious beliefs to which they may not belong themselves.

Designing Microfluidic Devices to Evaluate Interactions of Circulating Tumor and Immune Cells with the Irradiated Vasculature —

Marjan Rafat, Chemical and Biomolecular Engineering, Biomedical Engineering

Despite the high risk of recurrence among TNBC patients, the steps of tumor cell reseeding and the roles of neutrophils and radiation-induced vascular damage remain unclear. Traditional in vitro models for studying mechanisms of cancer recurrence and metastasis (e.g., cell culture plates, organoids) fail to recapitulate the structure of vascular networks and do not incorporate flow, which

is a key component of cell movement and transport of nutrients and secreted factors through the vasculature. The Rafat Lab is interested in establishing a more biologically relevant model of the mammary tissue vasculature in order to study the mechanisms of TNBC recurrence after radiation. An REU student in the Rafat Lab will have the opportunity to participate in experiments related to the to the fabrication and application of a two-channel microfluidic device to determine the impact of irradiated vasculature on neutrophil infiltration and tumor cell recruitment. The necessary platforms and protocols are well-established in the Rafat Lab. The REU students will be trained to independently conduct experiments. The project is best suited for a student interested in learning about cell-cell interactions in the tissue and tumor microenvironment through the design of biomimetic devices.

Researcher Interests and Skills: The Rafat Lab is accepting undergraduate students who would like to conduct research at the interface of engineering and cancer biology.

Development of New pH-reactive Polymers for Drug Delivery —

Craig Duvall, Biomedical Engineering

The Duvall lab develops and tests novel pH-responsive, "smart" polymeric carriers to be formulated as micelles and/or polymer drug conjugates for pharmaceutical applications. There are currently no mainstream clinical drugs consisting of intracellular-acting biologic macromolecules (i.e. peptides, proteins, and nucleic acids). These classes of molecules are too large and polar to diffuse across cell membranes, and if taken up by endosomal pathways, the predominant fates are lysosomal degradation or exocytosis. The pH-responsive polymers developed in the Duvall lab are optimized to respond to the discrete pH difference between the extracellular and endosomal environments to trigger biomacromolecular drug endosomal escape and cytoplasmic delivery. Current applications range cancer and regenerative nanomedicine. For the summer project, the student will implement microfluidic fabrication methods to form nanoparticles optimized for tumor penetration and anti-cancer therapeutic efficacy.

Researcher Interests and Skills: This project is best suited to students interested in nanomedicine. The summer project will expose students to living free radical polymerization, polymeric nanoparticle fabrication and characterization, and basic cell culture and molecular biology techniques.

Development of New ROS-reactive Polymers for Drug Delivery —

Craig Duvall, Biomedical Engineering

The Duvall lab develops various "on demand" drug delivery systems that release drug or undergo phase changes based on environmental cues. Reactive oxygen species (ROS) are elevated, causing

"oxidative stress", in many chronic inflammatory diseases. We are developing new classes of polymeric nanoparticles, microparticles, and hydrogels with ROS responsive/scavenging properties. These systems are tuned to release drug "on demand" - i.e., when ROS levels become elevated, the delivery system is triggered to release anti-oxidant drug. Many of these systems are also engineered to have inherent ability to sequester ROS from the local environment, which is therapeutically useful in order to limit inflammation-associated host tissue damage and promote disease resolution. Current applications range wound healing, breast cancer metastasis, and arthritis therapies. For the summer project, the student will learn methods for synthesis of ROS-reactive polymers, creation and characterization of particle formulations, and measurement of ROS scavenging activity and downstream therapeutic benefits.

Researcher Interests and Skills: This project is best suited to students interested in polymeric biomaterials and inflammation. The summer project will expose students to living free radical polymerization, polymeric nanoparticle fabrication and characterization, and basic cell culture and molecular biology techniques.

Dynamic Metamaterials —

Jason Valentine, Mechanical Engineering

The Valentine group is focused on developing nanostructured materials, referred to as metamaterials, that have unique optical properties which arise from the structuring employed. The goal for this REU is the development of dynamic metamaterials that can have their optical properties adjusted in real-time using electrical, optical, or thermal modulation. The potential uses of these metamaterials are far ranging and include tunable optical elements such as lenses, optical displays, and tunable light sources. The REU student will work closely with the PI and a graduate student on this project and be responsible for fabricating and optically characterizing metamaterials with an emphasis on studying the time-varying properties including changes in transmission and reflection as well as the modulation speed.

Researcher Interests and Skills: This project is best suited for individuals with interests in nanoscale optics and materials, nanoscale fabrication, and experimental optics techniques. The individual should have completed an electromagnetics course and experience with experimental optics and spectroscopy would be beneficial.

Enabling Technologies for Biomanufacturing Extracellular Vesicle-Based Therapeutics —

Jamey Young, Chemical and Biomolecular Engineering

Extracellular vesicles (EVs) are cell-made particles that provide a natural mechanism of information and material transfer between cells. There is growing interest in large-scale production of EVs due to their potential use as carriers of drugs or other biomolecules to diverse cellular targets; however, the technology to mass-produce purified EVs with tunable and well-defined properties is still in its infancy. REU students will be recruited to work with senior members of our NSF Future Manufacturing project team to develop, test, and scale-up a flexible workflow for producing EVs from differentiated human induced pluripotent stem cells (iPSCs). Various biomaterials and cell encapsulation strategies will be examined for adapting adherent host cells to suspension culture, in order to maximize the volumetric rate of EV production in stirred bioreactors. In addition, REU students will contribute to the development of instructional "BioFab" modules that introduce digital fabrication and cell-based manufacturing concepts to Vanderbilt undergraduate students and high school students in the Metro Nashville Public Schools district.

Researcher Interests and Skills: This project is best suited for students interested in working at the interface between biomaterials and cell culture. They must also be interested in working in a team to develop instructional modules for undergraduate and high school students. The project involves a collaboration with the labs of Ethan Lippmann and John Wilson in Engineering, and Alissa Weaver in Cell and Developmental Biology. Development of "BioFab" modules will be led by Prof. David Florian.

Engineered Models of Vascularized Human Brain —

Ethan Lippmann, Chemical and Biomolecular Engineering

This project focuses on developing and validating engineered models of the human brain. The Lippmann Lab has longstanding expertise integrating human stem cell-derived neural and vascular cells with biomaterials and engineered microdevices to build representative models of vascularized brain tissue. Cell phenotypes in biomaterials and devices are robustly characterized to see how closely they approximate native human brain, while devices are also continuously optimized to support these cell phenotypes and impart different stimuli.

Researcher Interests and Skills: The REU student will work with graduate students or postdoctoral fellows to test new cell culture devices and characterize cell phenotypes in these devices. No prior experience is needed, but the student will be expected to work effectively with his/her hands and be open to trial-and-error strategies for system optimization.

Engineering Metasurfaces as a Thermal Radiation Source —

Josh Caldwell, Mechanical Engineering

Controllable spectral response and directionality of light from radiation sources are of critical interest within the mid-infrared (IR) range for waste heat recovery in energy storage, transportation, sustainability, as well as in chemical sensing, and imaging. Yet, the sources in the mid-IR are inefficient, bulky, expensive, and/or complex to design. On the other hand, metasurfaces - flat structured surfaces engineered to modify optical and electric functions - are promising platforms for tailoring optical properties. This allows thermal sources (e.g., old incandescent light bulbs) to be used as narrow band mid-IR radiation source with only heat as the input. Its compactness and simplicity provide significant advantages over the existing infrared light sources. In addition, highly anisotropic materials with exotic optical features in mid-infrared range provide viable candidates as a material for constructing such metasurfaces. An REU student working on this project will have the opportunity to take part in designing thermal metasurface devices, experience fabrication of the device using nanoscale lithography tools and nanoscale characterization of structure, optical response, and thermal behaviors. This will be achieved using tools such as Fourier transform infrared spectroscopy and nano-optical probes such as scattering-type scanning near-field optical microscopy, along with imaging microscopes such as scanning electron microscopes (SEMs) and atomic-force microscopes (AFM). The REU student will perform simulations for thermal design of the experimental setup and measure temperature of the device with thermocouples.of biomaterials.

Researcher Interests and Skills: This project is best suited for a student interested in heat transfer, nanofabrication, and material characterization. Student must have completed at least one semester of thermodynamics and two semesters of a laboratory course.

Intelligent Soft Miniature Robots Integrated with Flexible Sensors —

Xiaoguang Dong, Mechanical Engineering

Small-scale robots with an overall size less than one centimeter that can be wirelessly actuated, monitored and controlled, could revolutionize minimally invasive medical operations by allowing access to enclosed small spaces inside the human body and performing medical operations such as drug delivery, onsite biofluid pumping and biopsy. Wirelessly powered small-scale robots using stimuli-responsive material and mechanisms which can be actuated by magnetic fields are especially promising, as magnetic fields can penetrate most nonmagnetic materials such as biological tissue and induce relatively large magnetic forces and torques on the robot body for remote and precise actuation. Despite recent advances in this field, critical challenges still exist in creating intelligent miniature robots that could navigate through complex confined environments and demonstrate medical functionalities such as point-of-care diagnosis. This REU project aims at developing wireless soft robots for sensing tissue physiological properties. The REU student will work together with other lab members on the fabrication and test of flexible sensors based on various stimuli-responsive materials which will be integrated on miniature soft robots. Project outcomes include prototypes of the devices and a project report which could potentially be turned

into a manuscript to be submitted to a proper journal or a top robotic conference such as RSS, ICRA, etc.

Researcher Interests and Skills: This project is best suited for a student interested in soft materials, flexible electronics, 3D printing, and the applications of materials in robotics and medical devices. The student should have skills on mechanical design and soft materials.

Light in Quantum Materials

Richard Haglund, Physics and Astronomy

REU students working in the Haglund group will study the behavior of light in quantum materials with unusual optical and electronic functionalities, such as: vanadium dioxide nanostructures that transform from insulators to metals when irradiated by laser pulses; single-photon emission from two-dimensional crystals such as graphene-like hexagonal boron nitride; and optical harmonic generation in semiconductor-metal nano composites Students will be able to fabricate thin films, crystals or nanostructures of quantum materials, study nonlinear optical phenomena with femtosecond lasers or characterize single-photon emitters for quantum information applications. For example, a REU student in the summer of 2024 used photoluminescence spectromicroscopy to locate and select single-photon emitters on hexagonal boron nitride nano sheets.

Researcher Interest and Skills: This project is well suited for a student interested in nanoscale material synthesis, spectroscopy, optical physics or quantum optics. A sophomore-level course in modern physics or optics provides sufficient background. Experience with any of the following is a plus for this project: optical spectroscopy, lasers, LabView instrumentation and software, computer simulations and software (e.g., Python, Matlab, Mathematica or C++), material characterization (e.g., X-ray diffraction, Raman spectroscopy) and microscopy.

Molecular Design of Membranes for Polar Solvent Dehydration

G. Kane Jennings, Chemical and Biomolecular Engineering

Conventional thermal separations of water-solvent mixtures are energy intensive and costly. Aqueous polar solvent mixtures often present the challenge of azeotrope formation, requiring additional operations for separation. Membrane-based separations have emerged as an energy- and cost-efficient alternative separation method, with pervaporation as the prime candidate for separating these mixtures. Current polymeric pervaporation membranes are limited to a few commodity polymers that are often not molecularly designed for the intended separation, leading to low fluxes and mild selectivity. In addition, the synthesis of these polymers and fabrication of the membranes requires several liters of volatile organic solvent per gram of polymer deposited. This project combines polymer synthesis and deposition into one step to rapidly produce thin polymer

films with a wide variety of compositions through a process we term spin-coating ring-opening metathesis polymerization (scROMP). The scROMP approach allows for thin polymer films to be fabricated in under 3 min on coupon supports using cyclic olefin monomers, Grubbs 3rd generation catalyst, and as little as half a mL of solvent per 35 cm2 of polymer selective layer. The scROMP method is being paired with molecular simulations to sample the wide array of potential compositions and accelerate the discovery of new polymer membrane selective layers. Polymers fabricated through scROMP are characterized by ATR-FTIR, stylus profilometry, SEM, contact angles, and GPC to determine properties as scROMP parameters and monomers are varied. The student participating on this project will learn to fabricate thin film composite membranes using scROMP, characterize them with various methods noted above, and measure component fluxes and selectivities of water-polar solvent mixtures. The student will vary polymer side chain composition and cross-linking density to determine the effects on glass transition temperature and membrane separation performance.

Researcher Interests and Skills: This project is appropriate for a student who has an interest in investigating sustainable energy and in characterizing polymeric materials and surfaces. Students should have completed general and organic chemistry courses.

Nanophotonic Biosensors —

Sharon Weiss, Electrical and Computer Engineering, Physics

Accurate and reliable detection of small, low molecular weight molecules is a major challenge for current sensor technology. The detection of these species is critical for applications including identification of disease biomarkers. The Weiss group is investigating the use of nanoscale porous silicon optical structures as promising sensors for small molecule detection. Paper-based, smartphone-compatible sensor platforms for point-of-care diagnostics are being explored. An REU student in the Weiss group will have the opportunity to participate in experiments and calculations related to the design, fabrication, and characterization of optical biosensors. Necessary fabrication and measurement systems, as well as simulation infrastructure, are well-established in the Weiss lab. The REU student will be trained to independently conduct experiments. This project is best suited for a student interested in materials, chemistry, and optics.

Researcher Interests and Skills: The REU student must have completed at least one semester of physics and one semester of chemistry.

Nanoscale Optical Trapping —

Justus Ndukaife, Electrical Engineering

The Ndukaife Research group conducts research at the interface between nanophotonics and microfluidics to develop new lab-on-a-chip devices for trapping, manipulation, sorting and sensing of nanometric objects and biological molecules, which are too small to be trapped by the conventional optical tweezers that was recently recognized with a 2018 Physics Nobel Prize. For this particular project, we are interested in developing and characterizing novel optical nanotweezers that can trap and manipulate very small objects that are at least a thousand times smaller than the thickness of the human hair using light and electric voltage.

Researcher Interests and Skills: This project is best suited for a student interested in nanotechnology and optics. Students should have completed at least general chemistry with a corresponding lab course.

On-chip Silicon Nanophotonics —

Sharon Weiss, Electrical and Computer Engineering, Physics

Silicon has traditionally been associated with being the most favorable material platform for most modern microelectronics technologies due to its electronic properties, compatibility with lithographic patterning, and earth abundance. However, silicon is also a favorable material platform for supporting light propagation, and silicon photonics is now considered to be the leading platform to achieve faster data transfer speeds on chip and a promising platform for quantum information science. The Weiss group is investigating on-chip silicon photonic components with subwavelength features and extremely strong electric field enhancements, which may lead to the next generation of on-chip photonic devices with ultrafast modulation speed, low power, and small footprint. An REU student in the Weiss group will be trained to independently carry out both experiments and simulations related to the design and characterization of advanced silicon nanophotonic components.

Researcher Interests and Skills: This project is best suited for a student interested in optics and semiconductors. The REU student must have completed at least two semesters of physics. Knowledge of electromagnetics would be helpful, but not required.

Optical Metamaterials for Information Processing —

Jason Valentine, Mechanical Engineering

The Valentine group is focused on developing nanostructured materials with tailored electromagnetic properties at optical frequencies, specifically for applications in photovoltaics, detectors, and other more exotic devices such as invisibility cloaks. These nanostructured materials, referred to as metamaterials, can be engineered with unique optical properties due to the type of structuring and constituent materials employed. The purpose of this summer research experience is

to implement metamaterials as pre-filters for image processing systems. For instance, metamaterials can be designed to perform derivatives and act as spatial filters, off-loading these operations from the digital system. Ultimately, this allows the digital system to operate faster and consume less power while also taking advantage of the unique design freedoms associated with metamaterials. The REU student will work closely with the PI and a graduate student on this project and be responsible for designing and optically characterizing metamaterials for various image processing tasks.

Researcher Interests and Skills: This project is best suited for individuals with interests in nanoscale optics and materials, nanoscale fabrication, and experimental optics techniques. The individual should have completed an electromagnetics course and experience with machine learning systems will be beneficial.

Patterning Sacrificial Smart Materials to Make Artificial Arterioles —

Leon Bellan, Mechanical Engineering

In the human body, vascular resistance is regulated by non-capillary microvessels which autonomically contract or dilate in response to appropriate biochemical stimuli. This process is critical to ensuring appropriate local pressure-flow relationships in tissue such that metabolic needs are met. Engineered microvasculature, however, is currently unable to replicate this critical functionality. This is because engineered microvasculature currently does not incorporate circumferentially aligned smooth muscle cells (SMCs) capable of regulating vessel lumen diameter. The goal of this project is to use 3D printed sacrificial templates to pattern microvascular channel networks in hydrogels and culture SMCs on the channels walls, and then induce appropriate SMC behavior and architecture using the cyclic wall stretch that results from pulsatile flow in these compliant matrices. Cells will then be exposed to various stimuli and their ability to modulate vessel lumen diameter will be characterized. This interdisciplinary project will also provide participating students with valuable materials processing and cell culture skills, and expose students to the challenges and excitement of novel biomaterials micropatterning techniques. Students will work with the PI, and graduate students and postdoctoral researchers in the Bellan lab, as well as with collaborators in other departments at Vanderbilt University.

Researcher Interests and Skills: This project is appropriate for students who are interested in hands-on experimental work and would like to learn more about biomaterials, flow characterization in a biomedical context, and 3D printing.. Experience with programming, electronics, biomaterials, or cell culture techniques would be beneficial but is not required.

pH-Responsive Polymeric Nanocarriers for Delivery of Vaccines and Immunotherapies

John T. Wilson, Chemical and Biomolecular Engineering

The Wilson ImmunoEngineering Lab (www.immunoengineeringlab.org) focuses on the treatment and prevention of disease through the design of nanoscale drug delivery systems that modulate the immune system. A major focus of our work is the design of protein, polymer, and lipid-based nanocarriers to enhance the intracellular delivery of the key components of vaccines and cancer immunotherapies. The summer research project will focus on design, testing, and optimization of novel drug delivery platforms for modulation of the immune system. REU students working in the Wilson group have the opportunity to learn a diversity of skills and concepts, potentially including polymer synthesis, colloidal self-assembly, nanoparticle characterization, protein engineering, bioconjugate chemistry, and immunobiology.

Researcher Interests and Skills: This project is best suited for a student interested in materials synthesis and characterization and the applications of materials in biology and medicine. Student must have completed three semesters of a laboratory course.

Phase Control in the Synthesis of Metal Sulfide Nanocrystals

Janet Macdonald, Chemistry

Metal sulfide nanocrystals have been a cornerstone of nanotechnology research: as examples Quantum dots of CdS, PbS and ZnS; graphene-like MoS_2 and lithium ion storing cobalt sulfides. As we seek new nanoscale materials for their electronic, catalytic, thermoelectric and optical properties we need to mine farther and farther reaches of the periodic table. The geologic record tells us there are many untapped materials: there are nine known iron sulfides, four cobalt sulfides, seven nickel sulfides and ten copper sulfides. As physical and chemical properties depend on the identity and the specific arrangement on atoms in space, each of these phases has their own set of potentially revolutionary properties for any number of applications from display technology, catalysis, batteries, non-linear optics and even cancer treatment.

Some of the aforementioned metal sulfides have never been prepared as nanocrystals before, as to date, our syntheses have been mostly serendipitous. The community of chemists do not have the level of synthetic skill or knowledge of how to tweak a “failed” reaction to select for one crystal phase over another. In this project we will use libraries of organosulfur reagents to tease out how kinetics and decomposition mechanism each play a role in the phase selective synthesis of transition metal sulfides. Our goal is to synthetically traverse the full phase space of metal sulfides.

Researcher Interests and Skills: This REU will be conducting synthetic experiments at high temperatures using Schlenk techniques, and will learn to use and analyze their products with X-ray diffraction as a routine technique. Transmission electron microscopy on promising samples will be performed with the aid of a graduate student. Given the synthetic rigors of the project, organic chemistry with a laboratory component is a minimum requirement, or similar synthetic laboratory experience.

Prediction of Hybrid Thermal Modes for New Class of Energy Conversion Materials —

Greg Walker, Mechanical Engineering

Exciting new research indicates that infrared metamaterials can be used for directional emission, spectrally selective emission, and even possibly lasing. However, the design of these structures depends on light interaction with crystal lattice vibrational modes. Only through careful design and analysis can these novel materials be realized. We will use density functional theory and lattice dynamics modeling tools to understand phonon dispersions in highly structured materials to look for elusive but important hybrid modes, those where optical and acoustic modes overlap. These materials are important to a number of thermal energy devices such as thermophotovolatics, microelectronic optical interconnects, and infrared power transmission with a meaningful impact on energy utilization and storage technologies.

Researcher Interests and Skills: This project is best suited for students with an interest in using computers to solve engineering problems and an interest in learning about energy transport.

Probing Protein-Nanoparticle Interactions with Ultrafast Spectroscopy —

Lauren Buchanan, Chemistry

The Buchanan group focused on the use of 2D infrared spectroscopy to study protein structure and dynamics. In this project, we are interested in determining the structure of nanoparticle-bound proteins. For proteins, structure determines function; thus, for biomedical applications of nanomaterials to be viable, it is crucial to understand precisely how protein-nanoparticle interactions alter protein native structure. While it is well-established that global structural changes do occur upon binding to a nanoparticle surface, no one has managed to localize structural rearrangements to specific regions of the protein or determine the mechanisms by which nanoparticles affect protein aggregation. Using 2D IR spectroscopy and isotope labeling of proteins, we can achieve residue-level structural resolution of these complex systems and observe transient species that are inaccessible by conventional methods.

Researcher Interests and Skills: This project is best suited for students interested in physical or bioanalytical chemistry. Students should have completed at least general chemistry with a corresponding lab course.

Super-Planckian Far-Field Radiation via Non-equilibrium Polaritons

Deyu Li, Mechanical Engineering

The Li group explores energy transport at nanoscale, which provides critical knowledge for various engineering applications including microelectronic device thermal management, novel energy converters, and powerful cooling technology for buildings. Currently we are pursuing super-efficient radiative cooling strategies via a type of novel energy carriers, phonon polaritons, which result from coupling between infrared photons and atomic vibrations in polar materials. Our recent research results indicate a strategy to efficiently launch phonon polaritons to a number density that is higher than the corresponding equilibrium value, which could lead to super-Planckian far-field radiation. The REU student will work with a Ph.D. student/post-doc to conduct experimental measurements of thermal transport mediated by non-equilibrium phonon polaritons and understand/demonstrate its impact on thermal radiation.

Researcher Interests and Skills: This project is best suited for a student who is interested in experimental work exploring energy transport through nanostructures. The student should have completed college physics courses and taken some engineering classes.

Synthetic Biomaterials for Regenerative Medicine

Scott Guelcher, Chemical and Biomolecular Engineering

The Guelcher group focuses on the design and development of synthetic biomaterials for soft and hard tissue engineering and drug delivery. Examples include delivery of biofilm dispersal agents to prevent infection and accelerate healing of contaminated bone wounds, and delivery of Wnt signaling inhibitors to enhance cutaneous wound healing. REU students working in the Guelcher group would learn the fundamentals of biomaterials synthesis and drug delivery, and perform experimental measurements using both in vitro and in vivo models. For example, previous undergraduate researchers in the Guelcher lab have measured the ability of novel biomaterials to support osteoblastic differentiation and the release kinetics of antibiotics from polymeric scaffolds. Each student initially works very closely with the PI and a graduate student or postdoctoral associate in the group, with the best students becoming essentially independent by the time they leave the lab.

Researcher Interests and Skills: This project is best suited for a student with interests in drug delivery and cell culture. The student should have completed a laboratory course, and courses in

organic chemistry and biology would be helpful. The student will work in an inter-disciplinary environment with chemical engineers and molecular biologists to develop new therapies for regeneration of bone and soft tissue.

Thermal Management of GaN-on-Diamond Devices —

Mona Ebrish, Electrical and Computer Engineering

Gallium Nitride (GaN) devices integrated with diamond substrates offer unmatched potential for high-power, high-frequency electronics due to diamond's exceptional thermal conductivity. However, managing and characterizing heat flow across the GaN–diamond interface remains a key challenge for reliable device operation.

In this project, the student will work with graduate students to investigate thermal management in GaN-on-diamond devices. The effort will include the **setup, alignment, and calibration of a state-of-the-art thermal imaging camera** for measuring device heating profiles during operation. Calibration will involve working with vendor-provided software and building scripts in **Python or LabVIEW** to automate data acquisition and improve spatial/temporal resolution. The objective is to generate reliable thermal maps of operating transistors, enabling correlation between device structure, interface quality, and heat dissipation pathways.

Depending on interest and skills, the REU student may focus on:

- **Experimental setup and calibration** of the thermal imaging system.
- **Software integration and automation** of measurement workflows.
- **Data analysis**, including extraction of thermal resistance and hot-spot characterization.

This project provides hands-on experience with both semiconductor device physics and modern instrumentation, while also contributing to the broader effort of improving thermal design in wide-bandgap electronics. All steps are novel and may lead to contributions in **peer-reviewed journals or conference proceedings**.

Researcher Interests and Skills:
This project is ideal for students interested in thermal management, microelectronics, and device characterization. Students should have a **basic understanding of semiconductor physics** (e.g., diodes, transistors) and some familiarity with **programming in Python or LabVIEW**. Previous exposure to cleanroom environments, microelectronics fabrication, or device testing is helpful but not required.

Understanding and Discovery of Novel Solid Electrolytes for Solid-State Batteries

De-en Jiang, Chemical and Biomolecular Engineering

The Jiang Group at Vanderbilt focuses on computational materials and chemical sciences, aiming to develop predictive models and design principles for functional materials and molecules that advance a sustainable society. Our current research spans first-principles studies of electrode materials and charge storage, ion dynamics of advanced electrolytes, surface chemistry of 2D materials, computational nanocatalysis, simulations of separation media and processes, machine learning force fields (MLFFs), and foundation models. A recent thrust of the group is the exploration of solid electrolytes capable of conducting Li^+ and Na^+ ions at rates comparable to liquid electrolytes. This project centers on computational modeling of novel amorphous halide and oxide electrolytes, which represent some of the most exciting frontiers in solid-state battery research.

REU students in our group will gain hands-on experience in:

- Building atomistic models of solid electrolytes
- Performing first-principles electronic structure calculations and training MLFFs
- Simulating ion diffusion in solids using MLFF-accelerated molecular dynamics.

Through this work, students will develop an appreciation for quantum-mechanical methods, machine learning force fields, materials modeling, solid electrolytes, and energy storage technologies, while contributing to research that addresses critical challenges in clean energy.

Researcher Interests and Skills: This project is best suited for a student interested in computational nanoscience, materials chemistry, and battery research. Knowledge of Linux operating system, solid-state chemistry, and quantum mechanics is a plus.

Using Photosystem I for Solar Energy Conversion

David Cliffel, Chemistry

In a collaboration between the Cliffel and Jennings Research groups, we investigate the use of Photosystem I (PSI) in a non-biological setting to efficiently convert solar energy into electricity or fuels. By taking this nano-machinery found in the process of photosynthesis and integrating with different electrode materials, we aim to both increase the efficiency and decrease the cost of modern solar conversion techniques. REU students working with the Cliffel and Jennings research

groups learn the fundamentals of surface modification, surface analysis, and electrochemistry. For example, previous REU students have worked on projects integrating and analyzing PSI with carbon-based electrodes, and measured the resulting photocurrents of these systems. Typically 2 students work closely with and graduate students in both the Cliffel and Jennings lab.

Researcher Interests and Skills: This project is best suited for a student interested in chemistry or chemical engineering. Students should have completed at least general chemistry with a corresponding lab course.

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